# MD. SAZZAD HOSSAIN

Khidramalanchi, Tomaltola, Bagatipara
Natore 6410, Bangladesh
Date of birth: 25 Jun 2003
Portfolio: md-sazzad-eee.github.io

Phone: (+880) 01303337052 (Mobile, WhatsApp)
Email: md.sazzad.eee@gmail.com
LinkedIn: MD. SAZZAD HOSSAIN
GitHub: md-sazzad-eee

## RESEARCH

**Inverter control and stability:** Control and stability of grid-forming and grid-following inverters in weak, low-inertia grids, including adaptive mode switching and stability-aware demand scheduling.

**Digital twins:** Physics-informed, self-calibrating digital twins of inverters and batteries that track parameter drift, with identifiability analysis and safe excitation design for calibration.

**Trustworthy machine learning:** Uncertainty-aware machine learning with conformal guarantees for power-system monitoring and detection of cyber-physical attacks.

**Power-electronics reliability:** Electrothermal modeling and condition monitoring of SiC and GaN power electronics, including junction-temperature estimation and component health.

## EDUCATION

**American International University-Bangladesh** — Dhaka, Bangladesh
**B.Sc. in Electrical and Electronic Engineering** — Jan 2022 to Dec 2025
*Final grade: 3.57/4.00 | Credits: 148*

## RESEARCH EXPERIENCE

**Research Centre for Green Energy, Electro-Anatomy, and Digital Intelligence (ReGED)**
**Research Assistant**, Dhaka, Bangladesh — Feb 2026 to Present

- Build physics-based and data-driven digital twins of PV inverters, grid-forming batteries and SiC power modules in MATLAB/Simulink and COMSOL.
- Study identifiability, Fisher-information excitation design and online calibration of inverter digital twins.
- Develop uncertainty-aware detectors with conformal guarantees for cyber-physical attacks on power systems.
- Prepare journal and conference manuscripts with the group, from modeling and simulation to writing and revision.

## JOURNAL

**Under review**

1. **Md. Sazzad Hossain**, Apurba Joty Roy, Abu Shufian, Debashish Kumar Ghosh, Md. Sajid Hossain, "*Digital-Twin Staleness Detection in Renewable Generation: A False-Alarm-Matched Benchmark of Sequential Drift Detectors across Sixteen European Fleets*," Next Sustainability (Elsevier), manuscript NXSUST-D-26-01704.
2. **Md. Sazzad Hossain**, Abu Shufian, Md. Sajid Hossain, Md. Rifat Hazari, Md. Saniat Rahman Zishan, "*Thermal-History Ambiguity in Grid-Connected SiC Photovoltaic Inverters: Electrical Observability, Physics-Constrained Estimation and Uncertainty-Aware Derating*," Scientific Reports (Springer Nature), under review.
3. **Md. Sazzad Hossain**, Apurba Joty Roy, Abu Shufian, "*Trustworthy False Data Injection Attack Detection Using a Conformalized Digital Twin in Cyber-Physical Power Systems*," Energy Conversion and Management: X (Elsevier), manuscript ECMX-D-26-02450.
4. "*Information-Aware and Safety-Constrained Excitation Design for Digital-Twin Calibration of Grid-Following Inverters with LCL Filters,*" Scientific Reports (Springer Nature), under review.

**Submitted**

1. "*Differential Calibration Drift Limits Photonic Bond-Health Sensing in SiC Power Modules: A Detectability Law and Ratiometric Two-Heater Self-Calibration,*" Nature Communications (Springer Nature), under consideration.
2. "*A Multi-Cell Physics-Based Digital Twin with Closed-Loop Kalman Filtering and Remaining-Useful-Life Prognosis for Lithium-Ion Battery State of Health,*" Energy Conversion and Management: X (Elsevier), manuscript ECMX-D-26-02617, with editor.

**In preparation**

1. "*Physics-Informed Machine-Learning Grid-Strength Estimation for Adaptive Grid-Forming/Grid-Following Energy Management of PV-BESS Plants*."

CONFERENCE

**Published**

1. Shaharier Kabir, **Md. Sazzad Hossain**, Md Sultan Mahamud, Tasnim Kabir, Abu Shufian, Muhammad Masud Karim, Saniat Rahman Zishan, "*A Physics-Informed Self-Calibrating Digital Twin with Conformal Uncertainty for Multi-Inverter PV Power Systems*," IEEE TENSYMP, Penang, Malaysia, 29 Jun to 1 Jul 2026.
2. Asif Iqbal, Sumit Hassan Eshan, Md Farukuzzaman, **Md Sazzad Hossain**, Azizul H Tuhin, Sadia Ahmed, Samiul Hasan, Orpyta Karmokar Aupy, "*Optimized Transport Layers for Enhanced Efficiency in Perovskite-Perovskite Tandem Solar Cells*," PEEIACON, Southeast University (SEU), Dhaka, Bangladesh, 4 to 5 Dec 2025.

**Accepted**

1. **Md. Sazzad Hossain**, Apurba Joty Roy, Abu Shufian, Md Aktar Hossain, Md. A. Hamid Howlader, "*A Stability-Aware Demand-Side Management (DSM) Framework for Grid-Forming Inverter Transient Stability Under EV Charging Disturbances*," IEEE ICICAIEECE 2027, UKF College of Engineering and Technology, Kollam, India, 21 to 22 Jan 2027 (Accepted). https://conference.ukfcet.ac.in/#home
2. "*Disturbance-Order Memory Indices and a Physics-Memory Digital Twin for Sequential Power System Disturbances*," IEEE PES International Meeting (PES-IM) 2027, Beijing, China, 10 to 13 Jan 2027, paper 27INT0957 (Accepted).
3. "*Session-Aware, Episode-Disjoint Detection of Rare Cyber-Physical Attacks in Midstream Pipeline SCADA with Calibrated Uncertainty and Physics-Consistent Explanations*," IEEE PES International Meeting (PES-IM) 2027, Beijing, China, 10 to 13 Jan 2027, paper 27INT0980 (Accepted).
4. "*An Event-Excited Digital Twin for Non-Invasive DC-Link Capacitor Health Estimation in Grid-Forming PV-Battery Inverters*," IEEE PES International Meeting (PES-IM) 2027, Beijing, China, 10 to 13 Jan 2027, paper 27INT0981 (Accepted).
5. "*Saturation-Aware Digital Twin of a VSG-Controlled Battery for Online Inertia and Damping Identification and Tamper Versus Degradation Discrimination*," IEEE PES International Meeting (PES-IM) 2027, Beijing, China, 10 to 13 Jan 2027, paper 27INT0984 (Accepted).
6. "*Electrothermal Validity Boundary of a Fixed-Parameter Digital Twin of the LCL Inductor in PV Inverters During Voltage Sags*," IEEE PES International Meeting (PES-IM) 2027, Beijing, China, 10 to 13 Jan 2027, paper 27INT1016 (Accepted).

**Under review**

1. **Md. Sazzad Hossain**, Apurba Joty Roy, Abu Shufian, Raihan Chowdhury, Goro Fujita, "*Calibrated Staleness Detection for Solar Digital Twins and the Dominance of the Retraining Window over the Detection Trigger*," NSysS, Cox's Bazar, Bangladesh. https://cse.buet.ac.bd/nsyss2026/
2. **Md. Sazzad Hossain**, Apurba Joty Roy, Abu Shufian, Raihan Chowdhury, Goro Fujita, "*A Multi-Cell Physics-Based Digital Twin with Closed-Loop Kalman Filtering and Remaining-Useful-Life Prognosis for Lithium-Ion Battery State of Health*," NSysS, Cox's Bazar, Bangladesh. https://cse.buet.ac.bd/nsyss2026/
3. "*Counterfactual Abduction for Shift-Robust Intrusion Detection in Midstream Liquid Pipeline SCADA Under Setpoint Changes*," IEEE PES International Meeting (PES-IM) 2027, Beijing, China, paper 27INT0873.
4. "*Loop-Integrity Parity: Zero-Shot Detection, Localization, and Signal Reconstruction of Cyber-Physical Attacks in a Midstream Pipeline Control Loop*," IEEE PES International Meeting (PES-IM) 2027, Beijing, China, paper 27INT0983.

**Submitted**

1. **Md. Sazzad Hossain**, Apurba Joty Roy, Abu Shufian, Raihan Chowdhury, Goro Fujita, "*Inverse Hot-Spot and Power Estimation in GaN HEMTs Using a Green's Function-Augmented Physics-Informed Neural Network*," ICECE 2026, Dhaka, Bangladesh. https://icece.org.bd/2026/
2. "*Quantum-Assisted Self-Reconfiguration of Grid-Forming Renewable Inverters Under Thermal and Grid-Strength Constraints*," ICREST 2027, paper 252.
3. "*Information-Driven Experiment Selection for Self-Calibrating Digital Twins of Grid-Connected PV Inverters, with Weak-Grid Identifiability Maps and a QAOA Benchmark*," ICREST 2027, paper 254.
4. "*Quantum-Assisted Electrothermal Operating-Point Optimization of SiC Photovoltaic Inverters Using*

*a Physics-Informed Digital Twin*," ICREST 2027, paper 255.

5. "*Drive-Cycle Validation of an Adaptive Unscented Kalman Filter for Joint SOC, Internal Resistance, and SOH Estimation in Electric Vehicle Batteries*," ICREST 2027, paper 256.
6. "*Plant-Anchored Statistical Stability Guarantees for Grid-Forming Inverters Using a Physics-Informed Neural Zubov Score*," IEEE TPEC 2027, Texas, USA, paper 111.
7. "*Identifiability Audit of Physics-Informed RUL Estimation and Image-Based Fault Diagnosis for PV Digital Twins*," ICREST 2027, paper 422.

## PROJECTS

**AI-Based Motion Detection Edge Computing System** Apr 2024 to May 2025

- Designed an intelligent edge computing system that captures and processes images using computer vision algorithms to detect motion through sequential frame analysis.
- Implemented AI-based classification on captured images to identify motion events and trigger automated responses.
- Integrated an AI camera module with Raspberry Pi 5 and a vision kit for complete on-device processing without cloud dependency.
- Optimized the image-capture pipeline and motion-detection algorithms for low-latency performance on edge hardware.

## WORK

**Energypac Engineering Ltd.** Dhaka, Bangladesh
**Internship Trainee** Jun 2025 to Jul 2025

- Gained hands-on experience in the manufacturing and testing of power equipment including transformers, switchgear, and protective devices.
- Assisted with equipment assembly, relay settings, and quality inspections.
- Participated in high-voltage testing procedures in accordance with IEC standards.
- Supported testing and commissioning activities for power distribution equipment.

## SKILLS

**Power Systems & Renewable Energy:** MATLAB & Simulink, PSCAD, PSS/E, DIgSILENT PowerFactory, HOMER Energy, RETScreen

**Multiphysics, FEM & EM Simulation:** COMSOL Multiphysics, CST Microwave Studio, Antenna Design, OptiSystem

**Programming & Machine Learning:** Python, C, C++, NumPy, Pandas, SciPy, Scikit-learn, PyTorch, TensorFlow, OpenCV, Matplotlib, Seaborn

**Circuits, PCB & IC Design:** KiCad, EasyEDA, NI Multisim, Proteus, Cadence Virtuoso

**Mechanical & Electrical CAD:** Autodesk Fusion 360, SOLIDWORKS, FreeCAD, AutoCAD, AutoCAD Electrical

**Embedded Systems:** Raspberry Pi 5, ESP32, Arduino Uno, AI Vision Kit, Motion Sensors

**Tools, Documentation & Design:** Git/GitHub, Visual Studio Code, Google Colab, LaTeX (Intermediate), Overleaf, Microsoft Office, Inkscape, CorelDRAW

**Professional Skills:** Research Skills, Problem Solving, Team Leadership, Project Management, Presentations, Marketing

## LANGUAGES

**Mother tongue:** Bengali
**Other languages:** English, Hindi, Urdu